

Governance Readiness Gaps in Organizational AI Deployment: A Triangulated Analysis of Threats, Incidents, and Practice

Abstract

Organizations are scaling artificial intelligence (AI) with unprecedented ambition, yet governance readiness lags far behind deployment pace. Drawing on dynamic managerial capabilities theory and institutional decoupling, we investigate how *trust readiness* (governance capability) and *integration readiness* (architecture capability) shape the link between organizational AI orientation and realized AI value. We triangulate three publicly available secondary data sources — the MITRE ATLAS adversarial threat matrix (52 case studies), the AI Incident Database (1,362 incidents), and the U.S. Executive Order 13960 Federal AI Use Case Inventory (1,757 use cases) — through a cross-taxonomy mapping anchored to empirically derived constraints to AI scaling. Our analysis reveals governance theater: while 61% of federal AI deployments report internal review approval, fewer than 9% report substantive safeguards such as impact assessments, bias mitigation, or independent evaluation — and risk-tiering does not rescue governance depth. A use-case-level logistic regression shows that integration readiness is the dominant predictor of operational deployment (OR=1.40, $p<0.001$), whereas trust readiness shows no independent effect, indicating that architecture — not governance — currently binds AI value realization. Across 14 convergent experiments, commercial procurement emerges as the strongest structural barrier: vendor-supplied systems are 71% less likely to report impact assessment after controls (OR=0.29, $p<0.001$). We develop four propositions and one extension linking governance theater, architectural binding constraints, procurement opacity, and sector-calibrated governance to AI value realization. The study contributes to IS research by (1) documenting governance theater empirically, (2) identifying procurement opacity and missing architectural enforceability as structural antecedents, and (3) providing a replicable triangulation pipeline using public data.

Keywords: AI governance, governance theater, trust readiness, integration readiness, secondary data triangulation

Introduction

Artificial intelligence has shifted from boardroom aspiration to organizational imperative. McKinsey’s latest global survey reports that 88% of organizations now use AI in at least one business function, yet the majority remain in piloting stages and only about one-third have begun scaling (Singla et al., 2025). Over 50% of firms cite legacy IT architectures as the primary barrier (Bobier et al., 2025). The gap between AI ambition and AI realization is widening, not narrowing.

We investigate this gap through a governance lens. Using the U.S. Federal AI Use Case Inventory (1,757 deployments across 38 agencies), we find that 61% of AI systems report internal review approval — but fewer than 9% report substantive safeguards such as impact assessments, bias mitigation, or independent evaluation. Even among the 227 systems explicitly flagged as impacting rights or safety, where deep governance is mandatory by executive order, substantive safeguards reach only 16–23%. This steep “surface-to-substance” drop-off is not merely organizational immaturity; it constitutes *governance theater* — a form of institutional decoupling (Meyer & Rowan, 1977) in which organizations adopt formal governance structures while the operational routines needed to enact them remain absent.

Documenting governance theater raises a harder question: *why does it persist?* Organizations face no shortage of governance frameworks — NIST AI RMF (National Institute of Standards and Technology, 2023), the EU AI Act (European Parliament and Council of the European Union, 2024), MITRE ATLAS, and OWASP Top 10 for LLM Applications (OWASP Foundation, 2025) all prescribe risk-tiered, lifecycle-managed governance. Yet even under federal mandate, substantive safeguards remain rare. We argue that the explanation lies not in awareness deficits but in *capability constraints*: governance demands architectural hooks to be enforceable, and commercial procurement introduces opacity barriers that render those hooks inaccessible. The puzzle deepens when one considers that the federal inventory is itself a product of regulatory mandate — Executive Order 13960 requires agencies to inventory and govern their AI use cases. If governance theater persists even

where both mandates and classification requirements exist, the constraint must lie deeper than regulatory design.

We frame this argument through two theoretical lenses. Upper echelons theory (Hambrick & Mason, 1984) establishes that CIO centrality shapes strategic AI orientation (Li et al., 2021; Reil-Jerenz et al., 2026), but orientation alone does not produce value. Drawing on dynamic managerial capabilities (DMC) (Adner & Helfat, 2003; Hossain et al., 2025), we argue that converting orientation into realized value requires two capability bundles: *trust readiness* (TR) — the governance capability bundle encompassing risk management, compliance, adversarial threat modeling, and accountability — and *integration readiness* (IR) — the architecture capability bundle encompassing data pipelines, evaluation infrastructure, and deployment controls. These bundles should function as strategic complements (Milgrom & Roberts, 1990), yet our evidence suggests they are currently decoupled: architecturally well-provisioned systems are far more likely to reach production (73% deployment rate for high-IR vs. 33% for low-IR), while governance safeguards show no relationship to deployment outcomes. Structurally, commercial procurement creates transparency barriers that prevent organizations from executing substantive safeguards even when policies demand them.

To ground this argument empirically, we triangulate three publicly available secondary data sources not previously connected in IS research: MITRE ATLAS (52 adversarial case studies representing what *can* go wrong), the AI Incident Database (1,362 incidents representing what *has* gone wrong), and the EO 13960 Federal AI Use Case Inventory (1,757 deployments representing what organizations *are actually doing*). Mapping these sources through a cross-taxonomy bridging table anchored to CIO-reported constraints to AI scaling (Reil-Jerenz et al., 2026), we construct a triangulated evidence base that links threat landscapes, incident patterns, and governance practice.

Through this triangulated analysis of adversarial threats, real-world incidents, and reported governance practices, we demonstrate that risk-tiered AI governance frequently functions as governance theater. We identify procurement opacity and missing architectural enforceability as binding constraints that prevent substantive safeguards from being implemented at scale, and we provide a replicable analytical framework that other researchers can extend with public data.

Theoretical Background

Upper echelons theory posits that organizational outcomes reflect the cognitive orientations and attention patterns of top executives (Hambrick & Mason, 1984). In AI contexts, the CIO’s role has evolved from technology steward to strategic partner (Chawla et al., 2023), and CIO centrality — board access, reporting structure, agenda ownership — together with board AI awareness significantly influence a firm’s strategic AI orientation (Li et al., 2021). Practitioner evidence reinforces this view: Reil-Jerenz et al. (2026) frame the contemporary CIO mandate around strategy, speed, and scaled intelligence; Bobier et al. (2025) find that IT leads or co-leads generative AI initiatives at 86% of AI-leading companies versus only 54% at laggards; and Sánchez Reina (2025) identifies agility and risk mastery as the defining items on the CIO agenda.

Strategic orientation, however, is necessary but insufficient. The dynamic managerial capabilities framework (Adner & Helfat, 2003; Hossain et al., 2025) identifies managerial human capital, social capital, and cognition as micro-foundations of capability building, yet does not specify *which* capabilities matter for AI value realization. We complement DMC with a governance-as-capability perspective (Teece, 2007), conceptualizing governance not as a static compliance constraint but as a dynamic organizational capability — a bundle of routines enabling the firm to sense governance requirements, seize governance opportunities (e.g., building trust as competitive advantage), and reconfigure governance structures as regulatory and technological landscapes evolve.

We operationalize this framework through two capability bundles. **Trust readiness** (TR) encompasses risk management, compliance, adversarial threat modeling, incident response, and stakeholder accountability, grounded in normative frameworks including NIST AI RMF (National Institute of Standards and Technology, 2023), the EU AI Act (European Parliament and Council of the European Union, 2024), and the MITRE ATLAS adversarial threat taxonomy. TR spans a continuum of depth: at the surface level, or-

ganizations establish internal review boards and seek authorization to operate (ATO), achievable through procedural compliance alone. At the intermediate level, conducting impact assessments and real-world testing demands domain expertise and evaluation infrastructure. At the advanced level, independent evaluation, bias/disparity mitigation, and structured appeal processes require deep technical access, cross-functional coordination, and sustained investment. The distribution of federal AI systems across this continuum serves as the paper’s core empirical anchor.

Integration readiness (IR) is the architecture capability bundle encompassing data pipeline governance, evaluation infrastructure, deployment controls, and code/model accessibility. Concretely, IR includes data catalogs, documentation practices, provisioned infrastructure, source-code access, and component reuse — the organizational and technical conditions for deploying and sustaining AI systems at scale. Critically, IR supplies the *enforcement hooks* that make governance actionable: without telemetry infrastructure, audit logging is impossible; without evaluation pipelines, bias testing cannot be conducted; without code access, independent review is foreclosed. Bobier et al. (2025) find that over 50% of firms identify legacy IT architecture as the primary barrier to AI scaling, underscoring IR’s strategic importance as a dynamic capability rather than a fixed asset.

A persistent finding in institutional theory is that organizations adopt formal structures to signal conformity with institutional expectations while decoupling those structures from day-to-day operations (Meyer & Rowan, 1977). Bromley & Powell (2012) refine this insight by distinguishing *policy-practice decoupling* — where adopted policies are not implemented — from *means-ends decoupling* — where implemented practices fail to achieve their intended outcomes. Both forms are germane to AI governance: an organization may establish an internal review board without conducting substantive assessments (policy-practice), or it may perform impact assessments that do not trigger downstream mitigation (means-ends). Recent work extends this logic to AI ethics specifically: Ali et al. (2023) document how organizational structures create barriers between stated AI ethics commitments and operational engineering practice. We adopt the term *governance theater* to describe this decoupling pattern — the systematic adoption of authorization artifacts without the operational safeguards needed to make authorization meaningful. Our empirical analysis tests whether a large-scale federal AI inventory exhibits signatures of both decoupling forms, and identifies the structural mechanisms — procurement opacity, missing infrastructure, and capability bundling thresholds — that sustain the decoupling.

We theorize that trust readiness and integration readiness are strategic complements (Milgrom & Roberts, 1990): the marginal return to each increases when the other is present. Governance controls require architectural hooks to be *enforceable* — audit logging requires telemetry infrastructure, human override requires architectural approval gates, and risk assessment requires evaluation pipelines. Conversely, architectural patterns require governance boundaries to be *safe* — multi-agent orchestration without access governance creates excessive agency risk; retrieval-augmented generation without data governance creates embedding vulnerabilities. Crucially, we predict that this complementarity is *conditional on governance maturity*: when TR remains theatrical — surface compliance without operational substance — it cannot interact productively with IR. We therefore expect integration readiness to be the binding constraint in current deployments, with complementarity emerging only as governance deepens.

Methodology

We follow a convergent triangulation design (Creswell & Plano Clark, 2018) using three publicly available data sources. Table 1 summarizes each source.

Table 1. Data sources

Source	Records	Coverage	Licence
MITRE ATLAS v4.x	52 case studies; 16 tactics; 155 techniques; 35 mitigations	Adversarial AI threats	Apache 2.0

Source	Records	Coverage	Licence
AI Incident Database (AIID)	1,362 incidents; 6,681 reports	Real-world AI harms (CSET: 201 classified; GMF: 278 classified)	CC BY-SA 4.0
EO 13960 Federal AI Use Case Inventory	1,757 use cases; 38 agencies; 62 variables	Governance and integration practice	Public domain

MITRE ATLAS provides a structured taxonomy of adversarial tactics, techniques, and procedures targeting AI systems, modeled on the MITRE ATT&CK framework for cybersecurity. Each of its 52 case studies documents a real or demonstrated adversarial scenario, the multi-stage attack chain, and applicable mitigations. The AI Incident Database, maintained by the Responsible AI Collaborative, is a systematically curated repository of documented AI harms whose incidents may include structured classifications from the CSET taxonomy (harm types and sectors) and the GMF taxonomy (technical failure modes). The EO 13960 inventory is a government-mandated catalog of federal AI use cases created under Executive Order 13960 (December 2020); each record spans 62 variables covering agency, development stage, purpose, AI techniques, PII handling, infrastructure, review status, and a comprehensive set of governance safeguard indicators.

Cross-taxonomy mapping. Because the three data sources use distinct classification schemes, we construct a bridging table that maps elements from each source to a common set of organizing constructs, anchored to McKinsey’s empirically derived “constraints to AI scaling” — eight barriers cited by CIOs spanning talent gaps (31%), integration complexity (29%), security (26%), regulatory concerns (24%), data foundations (21%), use case clarity (18%), ROI measurement (17%), and change management (16%) (Reil-Jerenz et al., 2026). The mapping links 126 elements across all three sources via rule-based assignments with explicit rationale. Monte Carlo sensitivity analysis (1,000 iterations at 10–20% perturbation) confirms mapping stability (mean = 0.95). Importantly, the paper’s core findings — governance drop-off, IR dominance, and procurement opacity — are computed directly from EO 13960 variables and are not mediated by the cross-taxonomy map.

Pathway model. To provide a preliminary test of the theorized Orientation $\rightarrow$ TR $\times$ IR $\rightarrow$ Value pathway, we estimate nested logistic regressions at the use-case level ($n=1,757$) predicting operational deployment status. Trust readiness (TR, 0–9) counts nine governance safeguard indicators (ATO, internal review, impact assessment, real-world testing, independent evaluation, post-deployment monitoring, AI notice, disparity mitigation, appeal process). Integration readiness (IR, 0–7) counts seven architecture indicators (data catalog, documentation, custom code, source-code access, provisioned infrastructure, timely resources, code/data reuse). Orientation is an agency-level composite of portfolio size and domain breadth. Controls include development method and rights/safety classification. We estimate four nested models: (M1) controls only; (M2) + orientation; (M3) + TR + IR; (M4) + TR $\times$ IR interaction.

Exploratory analysis. We employ a Bayesian-surprise-based exploratory analysis engine that programmatically generates and tests hypotheses from the data schema. Across 400 experiments (Session 1: 100; Session 2: 300), 67.7% of hypotheses were contradicted — a result we treat as substantive evidence of governance gaps rather than methodological failure. Only findings supported by convergent multi-experiment evidence or independently confirmed through standard statistical tests are reported. All analysis code and data are available in the companion repository.

Findings

Our findings are exploratory, not confirmatory. We emphasize patterns supported by convergent evidence across multiple independent experiments or confirmed through standard statistical tests.

Governance Theater: Depth Collapse and Tiering Failure

Federal AI deployments exhibit high surface compliance but strikingly low substantive safeguards. Across 1,757 EO 13960 use cases, internal review is reported by 60.7% and Authorization to Operate (ATO) by 37.2%, yet substantive safeguards collapse to single digits: post-deployment monitoring (8.5%), real-world testing (8.5%), appeal process (8.1%), independent evaluation (6.8%), disparity/bias mitigation (5.9%), and impact assessment (5.2%). This ~55 percentage-point drop from internal review to the substantive safeguard floor reveals an environment in which review processes exist but the operational governance work that frameworks prescribe remains rare.

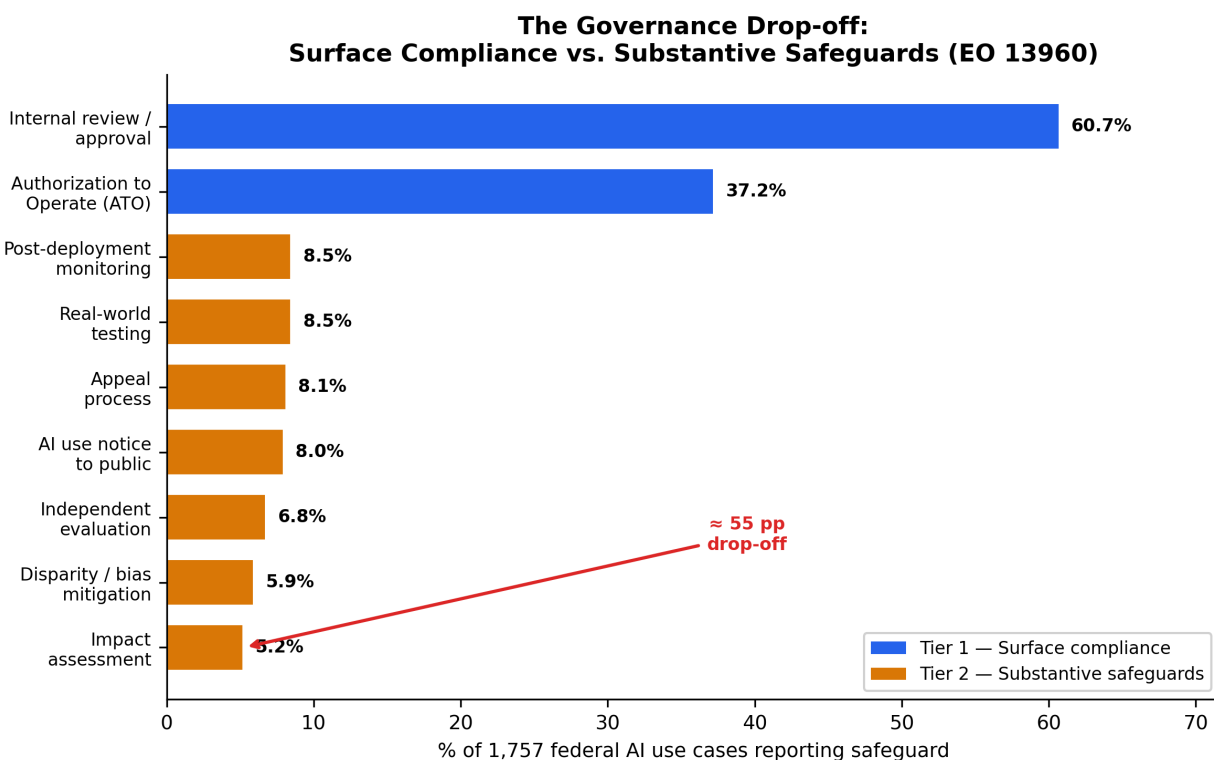


Figure 1: The Governance Drop-off: Surface Compliance vs. Substantive Safeguards (EO 13960)

This drop-off persists precisely where governance should intensify. Among the 227 use cases flagged as rights- or safety-impacting, deep safeguards rise but remain limited: impact assessment reaches 22.5%, independent evaluation 18.1%, and bias mitigation 16.3%. Even in this mandatory subset, the gap is attenuated rather than eliminated.

Risk-tiering — the foundational design principle of contemporary AI governance — fails to rescue governance depth. High-impact systems are no more likely to report independent evaluation (EXP_146, $n=1,718$, $p<0.0001$; contradicted), nor are rights-impacting systems more likely to report impact assessment (EXP_207, $n=1,757$; contradicted). Six additional experiments replicated this pattern, and three further experiments testing policy-timing effects found no measurable increase in bias mitigation (EXP_182, EXP_241, EXP_268). In institutional-theory terms, this constitutes policy-practice decoupling (Bromley & Powell, 2012; Meyer & Rowan, 1977): adopted classification policies do not translate into differentiated operational practice.

A preliminary pathway test reinforces this interpretation. In nested logistic regressions predicting operational deployment (52.4% of use cases), integration readiness emerges as the dominant capability predictor: each additional IR indicator increases the odds of deployment by approximately 40% (OR=1.40, 95% CI [1.33, 1.48], $p<0.001$). Trust readiness shows no independent effect ($\beta=0.085$, $p=0.287$), and the TR $\times$ IR in-

teraction is marginally negative ($=-0.028$, $p=0.052$), suggesting weak substitution rather than the theorized complementarity. The orientation proxy is negative ($OR=0.76$, $p<0.001$), pointing to an ambition–execution gap in which portfolio expansion outpaces deployment capacity. In practical terms, governance safeguards neither accelerate nor impede deployment — consistent with governance theater, where safeguards lacking operational substance should not predict outcomes.

Binding Constraints: Procurement Opacity and Architectural Enforceability

Commercial procurement is the strongest and most consistently supported predictor of governance under-implementation — not merely a correlate of low governance but a structural opacity barrier that prevents organizations from translating governance commitments into implementable safeguards.

Across 14 convergent experiments, vendor-supplied systems are associated with lower transparency and weaker governance. Vendor procurement predicts significantly lower code access (EXP_237, EXP_245, EXP_247 — all confirmed) and lower data documentation (EXP_174, EXP_284 — confirmed). The pattern extends to deeper safeguards, where vendor systems consistently trail in-house systems: impact assessment (6.4% vs. 10.3%), post-deployment monitoring (11.2% vs. 14.4%), and appeal process (10.6% vs. 13.7%).

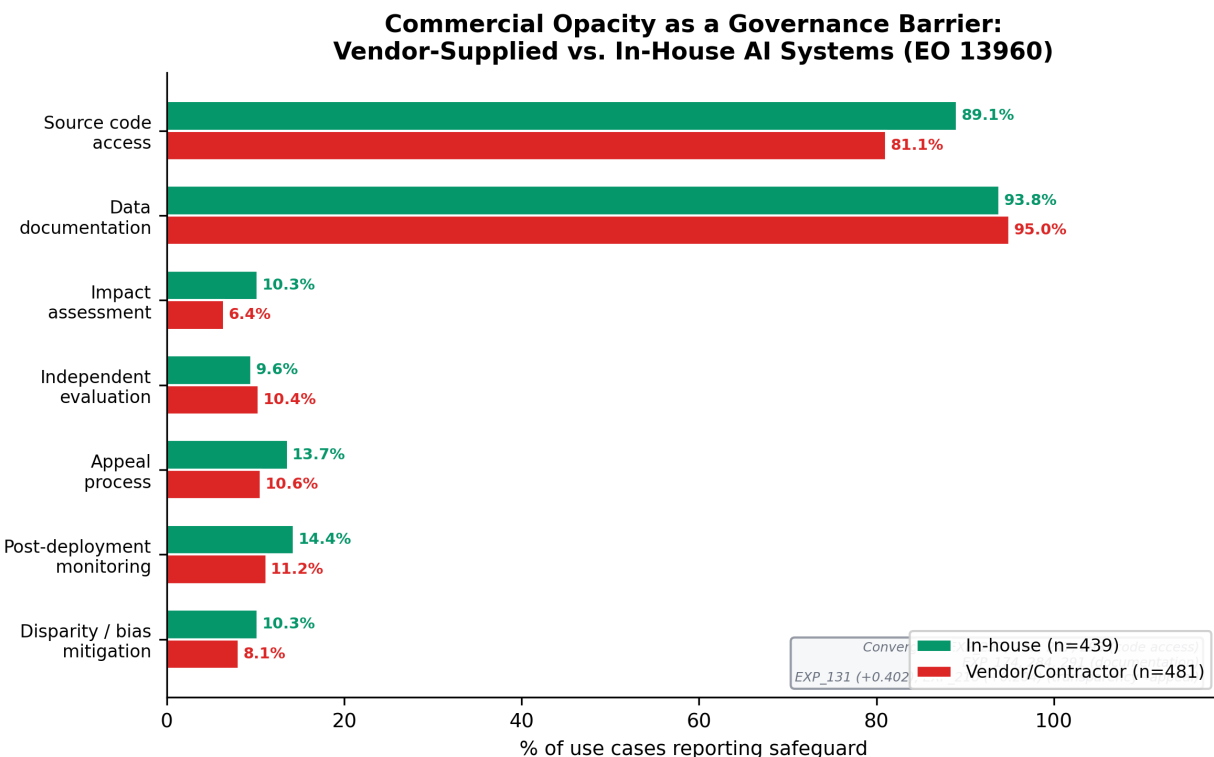


Figure 2: Commercial Opacity as a Governance Barrier: Vendor-Supplied vs. In-House AI Systems (EO 13960)

Multivariate logistic regressions ($n=1,107$) controlling for impact classification, public-facing status, operational stage, and agency fixed effects confirm these bivariate results. Impact assessment — the safeguard requiring the deepest vendor cooperation — shows the strongest effect: vendor-developed systems are 71% less likely to report it ($OR=0.29$, 95% CI $[0.15, 0.56]$, $p<0.001$). This effect *strengthens* with controls, indicating genuine procurement opacity rather than confounding. By contrast, independent evaluation, performable by external parties, shows no vendor effect ($OR=1.0$), confirming that the barrier is specific to activities requiring internal system access. Transparency deficits cascade into accountability: code access strongly predicts appeal processes ($OR=7.52$, $p<0.001$) and independent evaluation ($OR=19.48$, $p<0.001$).

Integration readiness provides the architectural context for these patterns. The IR index (mean=3.15, median=3) substantially outpaces the TR index (mean=1.49, median=1), and operational deployment rates differ sharply by IR level: 33% for low-IR (0–2), 57% for mid-IR (3–4), and 73% for high-IR (5–7). Where governance *is* present, it appears as coherent capability bundles rather than isolated controls: impact assessment predicts real-world testing (surprise +0.581), and high-IR systems are 9× more likely to exhibit bundled deep safeguards (14.9% vs. 1.6%, $\chi^2=87.3$, $p<0.001$; Spearman $\rho=0.228$, $p<0.001$). Yet assessments do not reliably trigger mitigation — a means-ends decoupling (Bromley & Powell, 2012) in which documentation artifacts function independently of operational action, consistent with the governance theater pattern.

Contextual Calibration: Sector-Specific Harm Fingerprints

AI harms are sector-structured rather than autonomy-driven, carrying important implications for governance design. Multiple experiments converge on sector-specific incident fingerprints: finance incidents skew toward economic harm, healthcare toward physical harm, and government and public-sector incidents toward civil rights harm (EXP_158, EXP_170, EXP_187, EXP_242, EXP_252 — all significant). Biometrics incidents show a disproportionate civil-rights association (EXP_168, $p<0.01$). Across technical failure types, fairness failures map to intangible harms while safety failures map to tangible harms ($\chi^2=12.97$, $p=0.0003$) — the most statistically robust cross-level pattern in the AIID analyses.

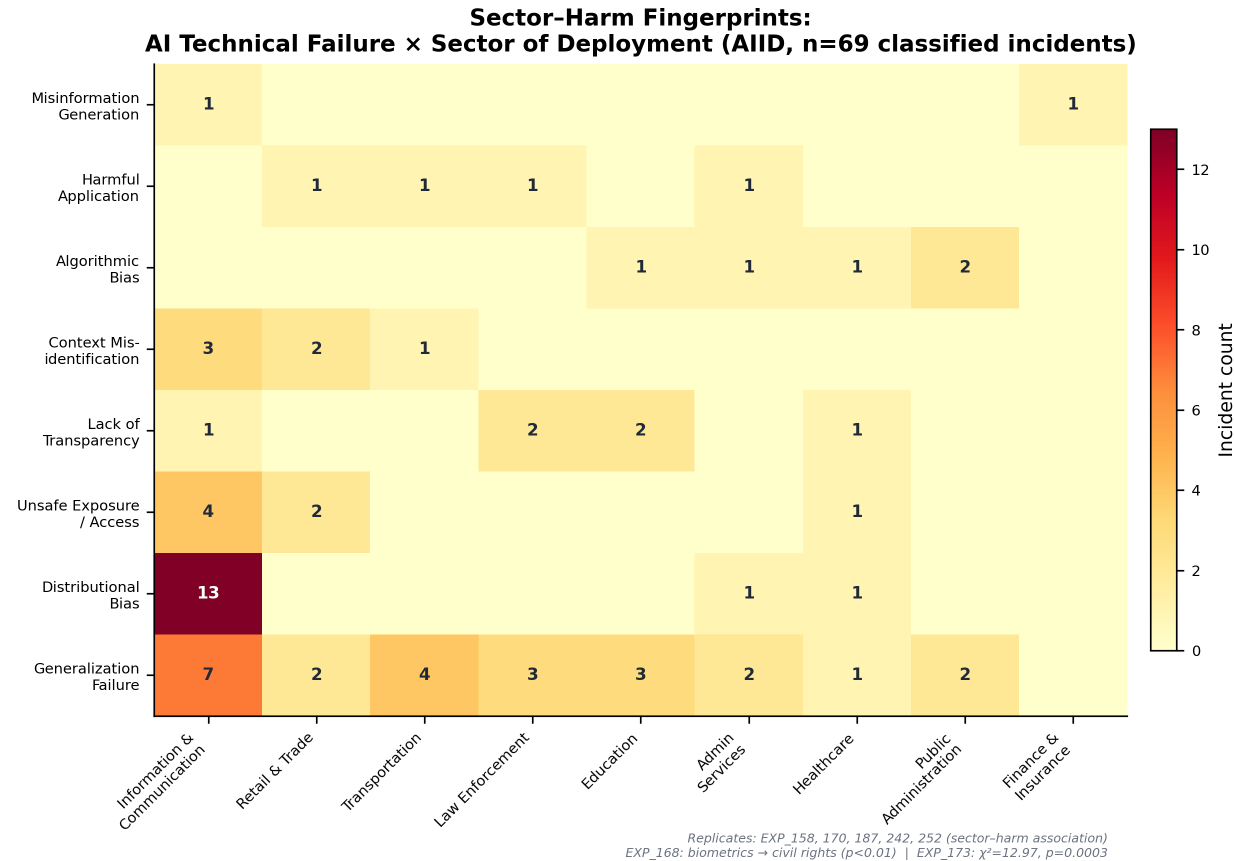


Figure 3: Sector–Failure Heatmap: Technical Failure Type × Sector of Deployment (AIID, n=135)

The incident landscape is also evolving temporally: post-2022 generative AI incidents have increased sharply in relative share, yet physical-world AI systems (robotics, autonomous vehicles) remain more severe per incident. The governance challenge therefore bifurcates into high-frequency, often intangible-harm GenAI failures (misinformation, discrimination) and lower-frequency but higher-severity physical safety failures. Notably, across 15 experiments the hypothesis “higher autonomy → more physical harm” was repeatedly

contradicted, cautioning against governance designs that treat autonomy as the primary risk stratifier while overlooking sector and modality effects that exhibit stronger empirical signal.

Cross-source triangulation reveals a threat–reality–practice divergence: ATLAS adversarial research emphases differ from AIID incident sector distributions (EXP_164; confirmed), and federal deployment sectors do not mirror incident concentration (EXP_108; +0.171). Governance programs derived primarily from adversarial threat research may therefore misallocate attention relative to incident reality. This reinforces the argument that trust readiness must be configurable — calibrated to empirically observed sector-specific harm patterns rather than applied uniformly from generic threat models.

Propositions

Based on the triangulated evidence, we develop four propositions and one extension.

Proposition 1 (CIO Centrality $\rightarrow$ AI Orientation). *The greater the CIO’s centrality in strategic decision-making, the stronger the organization’s strategic AI orientation.* Grounded in upper echelons theory (Hambrick & Mason, 1984) and supported by practitioner evidence (Li et al., 2021; Reil-Jerenz et al., 2026), P1 serves as the theoretical entry point: the pervasive pattern of governance underinvestment despite widespread deployment demonstrates that orientation cannot be inferred from deployment alone. P1 is testable via CIO survey designs and is not directly tested here.

Proposition 2 (Integration Readiness as Binding Constraint). *Trust readiness and integration readiness are strategic complements, but complementarity is contingent on governance maturity: only when trust readiness moves beyond surface compliance does the $TR \times IR$ interaction become positive.* IR is the dominant deployment predictor (OR=1.40, $p<0.001$) with TR showing no independent effect and the interaction marginally negative, indicating that complementarity remains aspirational at current maturity levels. High-IR systems are 9 $\times$ more likely to exhibit bundled deep safeguards, suggesting that the preconditions for complementarity exist in the high-IR tail but remain too rare to produce a population-level interaction.

Proposition 3 (Governance Theater under Regulatory Pressure). *Even under heightened regulatory pressure, risk-tiering regimes do not reliably produce higher deep safeguards; organizations exhibit governance theater in which surface compliance rises while substantive safeguards remain flat.* Mandates create demand for governance without supplying the capability infrastructure needed to fulfill it. P3 is falsified if risk tier or mandate strength robustly predicts deep safeguards, or if jurisdictions with enforcement mechanisms (e.g., EU AI Act penalties) show substantially higher substantive safeguard rates than mandate-only regimes.

Proposition 4 (Commercial Opacity Barrier). *Commercially procured AI reduces governance implementability by creating a transparency deficit that cascades into weaker accountability mechanisms and weaker safeguard bundles.* This is the strongest and most convergent empirical result (14 experiments, OR=0.29 for impact assessment), establishing procurement governance as a first-class theoretical object rather than a peripheral implementation detail: even motivated governance actors cannot execute independent evaluation or bias mitigation when system artifacts are inaccessible.

Extension E1 (Sector-Calibrated Governance). *AI harms exhibit sector-specific fingerprints, but governance investment does not align with incident reality; organizations require dynamic sensing capability to recalibrate governance priorities to sector-specific harm patterns rather than generic threat models.* This extension motivates why uniform governance fails and supports the configurability of the TR construct.

Discussion

TR/IR complementarity: aspiration versus reality. The pathway model provides a direct test of our central theoretical claim. IR dominates deployment prediction (OR=1.40, $p<0.001$) while TR shows no effect, and the $TR \times IR$ interaction is marginally negative. We interpret this as a maturity threshold effect: when mean TR is 1.49 out of 9, governance is too thin to interact meaningfully with architecture.

Complementarity thus characterizes the *target state* (deep governance paired with deep architecture) rather than the *current state* (theatrical governance alongside functional architecture). An alternative reading is organizational siloing: governance and architecture may be implemented by separate organizational units (compliance/legal vs. engineering/IT), producing independent — or even competing — investment patterns. In capability terms, the microfoundations of TR and IR have not yet been integrated into coordinated routines. Descriptive evidence supports the conditional interpretation: high-IR systems show significantly higher bundled safeguard prevalence (14.9% vs. 1.6%), indicating that the preconditions for TR×IR complementarity exist in the high-IR tail but remain too rare to generate a population-level interaction. Future cross-national or longitudinal designs in which governance maturity varies should find the interaction turning positive as mean TR increases.

Governance theater as capability failure. We characterize governance theater as the systematic decoupling of formal AI governance structures from the operational routines needed to enact them — an AI-specific instantiation of institutional decoupling (Bromley & Powell, 2012; Meyer & Rowan, 1977). Three reinforcing mechanisms sustain it. First, *procurement opacity*: vendor-supplied systems block access to model weights, training data lineage, and code needed for substantive safeguards. When the system boundary is vendor-controlled, even motivated governance actors cannot execute independent evaluation or bias mitigation, producing structural policy-practice decoupling. Second, *missing evaluation infrastructure*: the highest-demand adversarial mitigations (telemetry logging, adversarial input detection, model hardening) require monitoring pipelines that only 5–9% of federal systems report; without such infrastructure, governance policies cannot be operationalized and assessments cannot cascade into mitigation. Third, *capability bundling thresholds*: safeguards co-occur in coordinated clusters, suggesting that organizations must invest in complete bundles — assessment, testing, evaluation, and monitoring together — or produce no functional governance. This pattern echoes the DMC logic of coordinated sensing, seizing, and reconfiguring routines (Teece, 2007). Governance theater therefore signals *microfoundation deficit*, not merely motivational deficit — organizations may aspire to trustworthy AI but lack the measurement infrastructure and workflow gates needed to enact it.

Procurement opacity as structural barrier. The most novel empirical contribution is identifying procurement model as a boundary condition on governance capability. Vendor-developed systems are 71% less likely to report impact assessment even after controlling for mission type, agency, and impact classification (OR=0.29, $p<0.001$). This result extends governance-as-capability theory by revealing that trust readiness includes a *structural* dimension — the ability to access the artifacts required to execute governance processes. Even CIOs with ambitious AI orientation face discretion constraints imposed by vendor contracts and intellectual property claims. This dynamic echoes broader IS concerns about vendor lock-in and capability erosion (Ali et al., 2023), but in AI governance the consequences are sharper: opacity blocks auditability, contestability, and independent evaluation — core trustworthy AI obligations across NIST, EU, and ISO frameworks. The procurement finding also offers a plausible mechanism for why risk-tiering fails: if high-impact systems are disproportionately procured as vendor solutions, governance requirements cannot be implemented proportionally because the underlying artifacts are inaccessible.

Contradictions as evidence. The 67.7% contradiction rate in Session 2 strengthens rather than weakens the contribution. Contradictions cluster around core governance framework assumptions: risk-tiering should produce proportional safeguards, lifecycle maturity should strengthen governance, and autonomy should predict physical harm. The systematic falsification of these expectations demonstrates that contemporary governance frameworks encode aspirational capability models not yet routinized in practice. This positions contradictions as productive empirical objects in IS theory-building: governance aspiration has outpaced governance capability, and our triangulation pipeline pinpoints precisely where the divergence is greatest.

Implications, Limitations, and Conclusion

Research Implications

This study provides a replicable secondary-data framework for AI governance gap analysis, connecting “what can go wrong” (ATLAS), “what has gone wrong” (AIID), and “what is being done” (EO 13960) without

requiring proprietary organizational access. The finding that IR dominates deployment while TR shows no effect yields a testable conditional prediction: TR×IR complementarity should emerge as governance matures past a minimum threshold, testable through longitudinal tracking or cross-national comparisons of enforcement regimes (e.g., EU AI Act with penalty provisions versus mandate-only frameworks). More broadly, the work positions systematic contradictions as productive empirical objects in IS theory-building.

Practice Implications

For CIOs, the central implication is that governance must shift from checkbox compliance to bundle-based capability building. Organizations should implement safeguard clusters as standard architecture patterns — a V&V bundle (assessment → testing → evaluation) and an accountability bundle (assessment → notice → appeal). Procurement must become a first-class governance lever, embedding audit rights, documentation delivery, model transparency, and evaluation access requirements into AI vendor contracts. Finally, governance should be sector-calibrated rather than autonomy-calibrated, aligning safeguard priorities to empirically observed sector-specific harm fingerprints rather than generic threat models.

Limitations

The governance practice data is federal-only and U.S.-contextual, and the EO inventory is self-reported, introducing potential measurement noise. Pathway model variables are proxies (orientation as portfolio size/breadth; TR and IR as unweighted indicator counts) rather than validated scales. AIID structured taxonomies cover only 15–20% of the full corpus, although text-mined robustness checks replicate sector-harm associations on 2.9× the structured sample (Cramér’s $V=0.403$, $p<0.0001$). ATLAS comprises 52 case studies, limiting fine-grained subgroup analysis. The exploratory experiments should be validated with confirmatory designs, primary organizational data, and outcome measures beyond deployment stage.

Conclusion

This paper demonstrates that adversarial threat knowledge and real-world AI failures expose a persistent governance readiness gap in a large, policy-salient deployment portfolio. While 61% of federal AI systems report internal review, fewer than 9% report substantive safeguards, and integration readiness — not trust readiness — is the dominant predictor of operational deployment ($OR=1.40$). Commercial procurement emerges as a structural opacity barrier that undermines implementable governance ($OR=0.29$ for impact assessment). By triangulating threats, incidents, and governance practice, we offer a replicable method and a reconciled proposition set to guide IS research on how CIO-driven AI orientation can be converted into safe, scalable AI value.

References

- Adner, R., & Helfat, C. E. (2003). Corporate effects and dynamic managerial capabilities. *Strategic Management Journal*, 24(10), 1011–1025. <https://doi.org/10.1002/smj.331>
- Ali, S. J., Christin, A., Smart, A., & Katila, R. (2023). Walking the walk of AI ethics: Organizational challenges and the individualization of risk among ethics entrepreneurs. *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’23)*, 217–226. <https://doi.org/10.1145/3593013.3593990>
- Bobier, J.-F., Chatterjee, A., Ebeling, R., Grebe, M., Makarov, D., Milde, J., Nalgirkar, R., & Soininen, J. (2025). *The CIO’s role in AI value creation* [Industry Report]. Boston Consulting Group. <https://www.bcg.com/publications/2025/cios-role-in-ai-transformation-and-productivity>
- Bromley, P., & Powell, W. W. (2012). From smoke and mirrors to walking the talk: Decoupling in the contemporary world. *Academy of Management Annals*, 6(1), 483–530. <https://doi.org/10.1080/19416520.2012.684462>
- Chawla, R. N., Goyal, P., & Saxena, D. K. (2023). The role of CIO in digital transformation: An exploratory study. *Information Systems and e-Business Management*, 21(4), 797–835. <https://doi.org/10.1007/s10257-023-00651-1>

- Creswell, J. W., & Plano Clark, V. L. (2018). *Designing and conducting mixed methods research* (3rd ed.). SAGE. <https://us.sagepub.com/en-us/nam/designing-and-conducting-mixed-methods-research/book241842>
- European Parliament and Council of the European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act). In *Official Journal of the European Union: L series*. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- Hambrick, D. C., & Mason, P. A. (1984). Upper echelons: The organization as a reflection of its top managers. *Academy of Management Review*, 9(2), 193–206. <https://doi.org/10.2307/258434>
- Hossain, S., Fernando, M., & Akter, S. (2025). Digital leadership: Towards a dynamic managerial capability perspective of artificial intelligence-driven leader capabilities. *Journal of Leadership & Organizational Studies*, 32(2). <https://doi.org/10.1177/15480518251319624>
- Li, J., Li, M., Wang, X., & Thatcher, J. B. (2021). Strategic directions for AI: The role of CIOs and boards of directors. *MIS Quarterly*, 45(3), 1603–1644. <https://doi.org/10.25300/MISQ/2021/16523>
- Meyer, J. W., & Rowan, B. (1977). Institutionalized organizations: Formal structure as myth and ceremony. *American Journal of Sociology*, 83(2), 340–363. <https://doi.org/10.1086/226550>
- Milgrom, P., & Roberts, J. (1990). The economics of modern manufacturing: Technology, strategy, and organization. *American Economic Review*, 80(3), 511–528. <https://www.jstor.org/stable/2006681>
- National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.AI.100-1>
- OWASP Foundation. (2025). *OWASP top 10 for LLM applications*. OWASP. <https://owasp.org/www-project-top-10-for-large-language-model-applications/>
- Reil-Jerenz, A., Romanelli, G., Jogani, R., Catlin, T., Halawa, A., & Himatsingka, P. (2026). *The new CIO mandate: Strategy, speed, and scaled intelligence* [McKinsey Global Tech Agenda 2026]. McKinsey & Company. <https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-global-tech-agenda-2026>
- Sánchez Reina, D. (2025). *Strategic pivots will define the CIO’s agenda in 2026* [2026 CIO and Technology Executive Survey]. Gartner, Inc. <https://www.gartner.com/en/information-technology/insights/cio-agenda>
- Singla, A., Sukharevsky, A., Hall, B., Yee, L., Chui, M., & Balakrishnan, T. (2025). *The state of AI in 2025: Agents, innovation, and transformation* [McKinsey Global Survey]. McKinsey & Company. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>
- Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, 28(13), 1319–1350. <https://doi.org/10.1002/sm.j.640>